Do Ray-Traced Cities Make a Better Channel Prior for an Unseen Site?

A pre-registered study in simulation, and a test on measured channels that reverses it

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Abstract

Channel estimators for a newly deployed base station are usually built from 3GPP stochastic channel models, which describe no particular street. Ray tracers can simulate real city layouts instead. We ask whether a channel prior built from many ray-traced cities serves an *unseen* site better than one built from 3GPP CDL models, and whether learned estimators are what make the difference. Using Sionna RT on 14 city scenes (4 built-in, 10 generated from OpenStreetMap), linear MMSE estimation from covariance priors, and leave-one-city-out evaluation with outcomes and pass/fail bars written down before each main experiment, we find in simulation that a covariance prior pooled over many cities sits about 0.6 dB from a site-trained estimator with no site measurements—about what CDL reaches after 50 site measurements—and that pooling acts as insurance against a badly matched source city. Learned models added little: a CNN refiner gained at most 0.65 dB in channel estimation, and for channel prediction a simple per-user classical method beat CNN predictors by 1.8–12 dB at horizons of 2 ms and beyond; a CNN trained to correct the classical method added a repeatable 0.3–0.6 dB. We then tested the main simulation finding on measured channels from the DICHASUS outdoor dataset (1.272 GHz, three 2×8 arrays on one building). There, the pooled ray-traced prior was 3.2–4.1 dB *worse* than CDL on the full array with no site data. The simulated channels were cleaner in delay than the measured street; an exploratory rerun with diffuse scattering and diffraction enabled in the ray tracer brought the ray-traced prior to within 0.3–0.5 dB of CDL, but placing the simulated users at the real users' distances undid most of that (2.9 dB behind CDL): the measured street carries multipath the map-built scenes do not produce. All results are from one measured site and one band.

1 Introduction

A base station estimates the radio channel to each user from a few pilot symbols. Linear MMSE (LMMSE) estimation does this well when it knows the channel's second-order statistics (its covariance across antennas, subcarriers and time); learned estimators can in principle exploit more. At a new site, neither the covariance nor training data is available, so estimators start from a generic prior—typically the 3GPP CDL/TDL models of TR 38.901 [1]—and adapt as site measurements arrive.

Differentiable ray tracers such as Sionna RT [3] can simulate propagation in real city geometry. Recent work uses ray-traced data as a site-specific prior for a known site [7, 8, 9], pre-trains across source cities for CSI feedback [10], and compares stochastic with ray-traced data for machine learning [13]. We ask a narrower question: for a site *with no geometry model and no measurements*, is a prior pooled from *other* ray-traced cities better than the 3GPP models? And is a learned estimator what makes the difference?

We answer in three steps: (i) leave-one-city-out experiments in simulation, (ii) the same question for short-term channel prediction, where learning has more room, and (iii) a test of the main simulation finding on measured channels. Each main experiment's expected outcome and pass/fail bar was committed to the lab notebook before it ran; follow-up checks are labelled as post-hoc or exploratory. The code, notebook, logs and result tables are available with the paper.

2 Setup

Grid and estimation. A 5G NR-like OFDM slot: 3.5 GHz carrier, 30 kHz subcarrier spacing, 128 subcarriers × 14 symbols, DMRS-like pilots on every other subcarrier of symbols 2 and 11, and an 8-antenna half-wavelength uniform linear array at the base station. Channels are normalised to unit power per sample. The estimator is LMMSE from a covariance prior (R_{hp}, R_{pp}) estimated from 12k simulated channels. With N site measurements, the prior is shrunk toward their sample covariance with a weight chosen by 5-fold cross-validation. A residual CNN (zero-initialised last layer, so it starts at its LMMSE front end) tests whether anything beyond second-order statistics helps. Error is NMSE at 10 dB SNR unless stated, and the "ceiling" is LMMSE from 12k channels of the target site itself.

Channel sources. All channels are generated with Sionna [2]. *CDL*: 3GPP CDL-A to E with delay spread 30–300 ns and a random base-station heading per sample (a fixed heading made CDL artificially poor in an early run, which we corrected and report). *Ray-traced (RT)*: Sionna RT 2.2 with up to 3 specular bounces and refraction, in Sionna's four built-in scenes (Munich, Paris-Étoile, Florence, San Francisco) and ten New York State areas built from OpenStreetMap with geo2sigmap [4], reading buildings from an offline Geofabrik extract. Base stations are placed 25–40 m up near the scene centre and users within 250 m.

Protocol. Each city in turn is the unseen target; priors come from the other cities (pooled with equal weight) or from CDL. Test users are at base-station sites the priors never saw. The realism result was re-checked on fresh base-station sites and freshly drawn pre-training sets.

3 Results in simulation

Ray-traced vs CDL, four cities. With no site data, a prior pooled from the three other built-in cities beat CDL in Munich (+0.3–0.5 dB), Florence (+1.1–1.2) and San Francisco (+1.3–1.5) and lost in Étoile (–1.2 to –1.8) across the original and two fresh draws of sites and pre-training data. The advantage faded with site data: by N=1000 all priors were within ∼0.2 dB except Étoile.

Fourteen cities: pooling as insurance. Pooling k random other cities (8 draws each), the *worst* draw's gap to the ceiling shrank from 3.99 dB at k=1 to 0.82 dB at k=8, while the average went from 1.59 to 0.60 dB (Fig. 1): pooling protects against a badly matched source city (the worst single-city cases, 6–7 dB, all used a nearly empty open-area scene as the source for a dense downtown). The all-cities pool beat CDL in 11 of 14 targets by 0.29 dB on average. With no site data it sat 0.59 dB from the ceiling, about where CDL lands after 50 site measurements (0.47–0.73 dB across four draws of those 50). Once 50 measurements are blended in, the two are comparable (pooled ahead by ∼0.1 dB on built-up sites). Choosing a single best source city from 50 measurements worked with 3 candidates (mean regret 0.05 dB) but overfit with 15 (0.48 dB, against a pre-registered bar of 0.3).

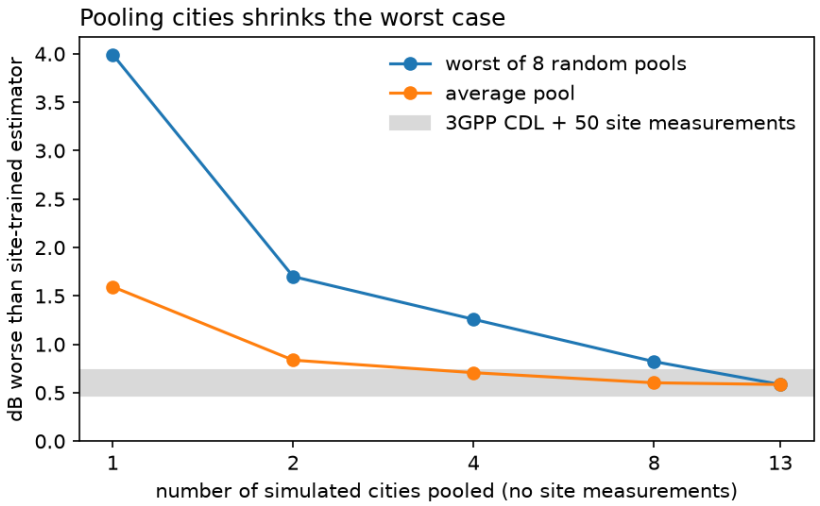


Figure 1: Simulation: gap to a site-trained estimator vs. the number of pooled ray-traced cities, no site measurements, mean over 14 targets. The grey band is 3GPP CDL after blending in 50 site measurements.

Learning in estimation. The CNN refiner added ∼0.2–0.45 dB on the CDL prior and often nothing on ray-traced priors, at most 0.65 dB in any cell (one seed). The covariance did almost all of the work.

4 Learning where it should have room: channel prediction

Predicting the channel 0.5–10 ms ahead from 1.4 ms of noisy pilot history (8 observations over 4 slots) is a task where per-user adaptation matters: a covariance pooled over users cannot know each user's Doppler. Sionna RT evolves a moving user by rotating the phases of a fixed set of paths; we checked that this matches re-tracing at the moved position up to ∼10 ms for nearly all users at 3.5 GHz (with the solver's deterministic mode), so we restricted horizons to 10 ms.

On 3GPP-compliant simulated channels, learned predictors have been reported to beat a Prony-based angle–delay baseline [11]. We compared repeating the last observation, space-time Wiener prediction from covariance priors, a classical per-user method ("tone": in the angle–delay domain, fit one Doppler tone per bin to the history and extrapolate), and CNN predictors in the antenna and angle–delay domains. The pre-registered test for "learning has room" failed: tone beat both CNN arms at every horizon from 2 ms in every city, by 1.8–4.7 dB in Munich, Étoile and Florence and by ∼9–12 dB in San Francisco (Fig. 2). The CNNs did beat covariance-based prediction by 2–6 dB at 2 ms; learning was competitive only one slot (0.5 ms) ahead.

A post-hoc follow-up trained a CNN to *correct* tone's prediction, with a loss weighted toward the predictable horizons. It added 0.3–0.6 dB at 2–10 ms in three of four cities (cells 0.26–0.65; two seeds agree within ∼0.2 dB) and cost 0.8–1.2 dB at 5–10 ms in San Francisco, where tone is already near-exact. Re-weighting the original CNNs' loss alone did not help. The caveat that matters most: fixed-path evolution is tone's own model, so on real channels tone would degrade by an amount this simulator cannot measure.

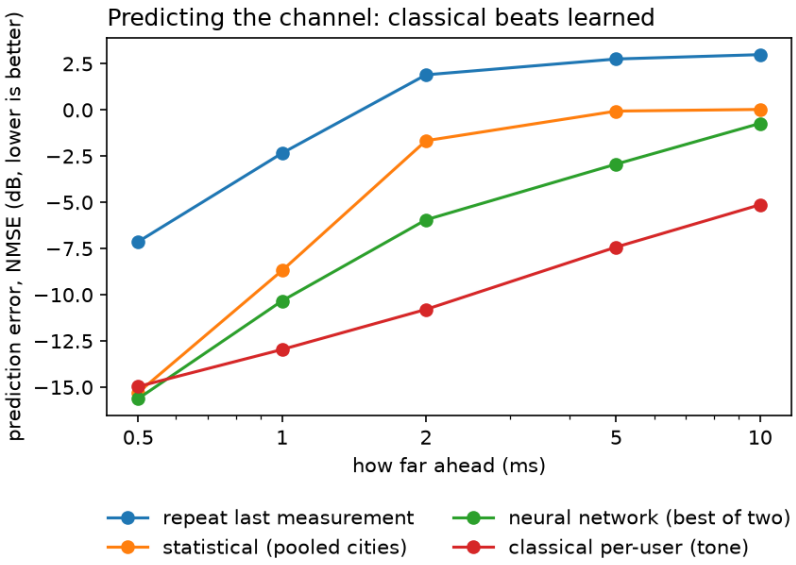


Figure 2: Simulation: prediction error vs. horizon, mean over four cities, 10 dB SNR, no site data. "Neural network" is the better of the two CNN arms in each cell.

5 Test on measured channels

Data. DICHASUS [5] outdoor set `dxxx` [6]: three 2×8 arrays ($\lambda/2$ at 1.272 GHz) on one building facade about 10 m up and tilted ∼20° down, a hand-pulled transmitter on the street, 50 MHz over 1024 subcarriers, tachymeter positions. Calibration files (also used for the ceiling), a site file (50 random measurements, 4 draws) and test files are disjoint recordings.

Processing. The receivers carry constant per-antenna phase and timing offsets, as the dataset authors describe; offset files are not shipped for this set. We fit, per antenna, a phase plus a delay relative to a reference antenna to the line-of-sight phase predicted from the tachymeter positions; the fit also identified the array layout (8-element axis horizontal: coherence 0.73 on array A versus at most 0.32 for the alternative layouts in an early fit; 0.80 on the full calibration set). Every source—measured, CDL and ray-traced—is generated on the full 50 MHz band and passes through one chain: align the first arrival to delay zero, keep a delay window of −200 ns to 2 μs, take the 128 subcarriers around the carrier (6.25 MHz), normalise. The grid is 16 antennas × 128 subcarriers in one snapshot, pilots on every other subcarrier. All 14 cities were re-simulated at 1.272 GHz with the 2×8 array, a directional 38.901 element pattern, a 20° downtilt and a facade-like placement (base station 10 m above ground, users 10–65 m in front, outdoor positions only, heights following terrain).

Results. The pre-registered claim that the pooled ray-traced prior beats CDL with no site data was falsified on all three arrays (Table 1, Fig. 3): it was 3.2–4.1 dB worse on the full array × frequency grid and 0.5–1.3 dB worse per antenna. On the full grid no single city beat CDL either. With 50 site measurements CDL still led by 1.2–1.7 dB on the full grid; per antenna both reached the site-trained level. Pooling still shrank the worst single-city case in 5 of 6 checks, but around a level below CDL. The pool sat 5.8–6.8 dB from the ceiling, against ∼0.6 dB in simulation.

array	LS interp.	CDL	RT pool (specular)	RT pool (scattering)	ceiling
A	−11.2	−17.9	−14.7	−17.6	−20.5
B	−11.2	−17.5	−13.4	−17.0	−20.2
C	−11.2	−17.9	−14.4	−17.5	−20.4

Table 1: Measured DICHASUS channels: NMSE (dB) at 10 dB SNR on the full 16-antenna × 128-subcarrier grid, no site measurements. Specular pool: 14 cities; scattering pool: 7 cities (exploratory).

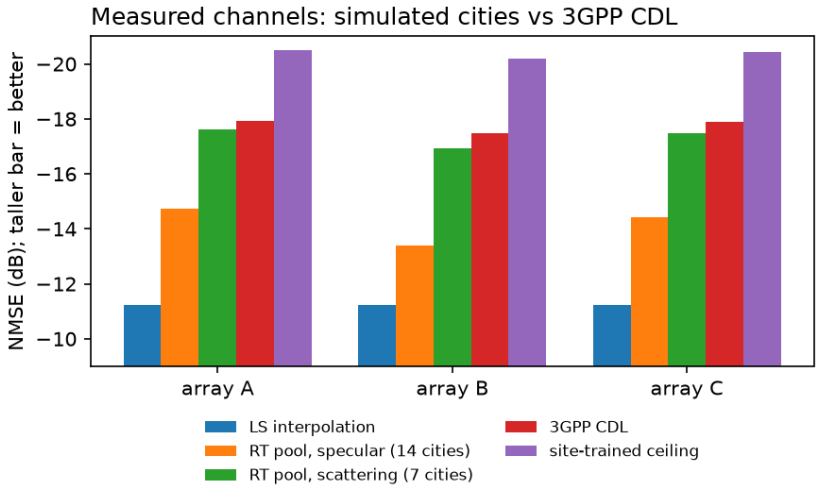


Figure 3: Measured channels, the same numbers as Table 1.

Why. The simulated channels were cleaner in delay than the measured street: 89–100% of energy in the first 160 ns tap across the 14 cities, against 79–84% measured (CDL 79%). Spatial concentration was not clearly different. The ray tracer ran with the original study's settings: specular reflection and refraction only, no diffuse scattering, no diffraction, every material's scattering coefficient zero. In an exploratory rerun, added after a dry run and labelled as such, we enabled diffuse scattering (coefficient 0.3, Lambertian) and Sionna's wedge diffraction for 7 cities. The pooled prior came within 0.3–0.5 dB of CDL on the full grid and level with it per antenna. A control with the same cities and the same number of base-station sites but no scattering attributes ∼2.8 dB of the ∼3 dB improvement on array A to scattering and diffraction and ∼0.2 dB to the extra sites.

Matching the users' distance made it worse. The simulated users sat farther from the array than the real ones (median 46 m vs 18–20 m). In a second exploratory probe, with scattering on and users placed 12–30 m away, the ray-traced prior fell to 2.9 dB behind CDL on array A (2.1–2.6 dB worse than the farther placement on every array and grid). Closer simulated users are more line-of-sight dominated (first-tap share 89–97%), while the real users at those distances show 79–84%: the measured street has multipath the map-built scenes do not produce, plausibly from objects they lack (vehicles, trees, street furniture, facade detail). The near-match of the scattering variant therefore came partly from placing users farther away than the real ones.

6 Discussion and limitations

The simulation finding was a home-ground effect, as the caveat written before the first run anticipated: test channels came from the same ray tracer as the priors. CDL is fitted to measurements; ray tracing at default settings encodes geometry but misses diffuse multipath, and every ray-traced candidate is biased the same way, so pooling cannot insure against it. Several limits apply.

- One measured site: one street, one building, three arrays on its facade, 1.272 GHz, mostly line-of-sight; LMMSE estimation only.
- The measured-channel test was pre-registered but not blind: a pilot had already scored CDL on the test data, and a dry run preceded the full run.
- The simulated placement was not matched to the real users (median 46 m and $\sim$10° below the array by the placement rule, versus 18–20 m and 24–26° measured), and one downtilt was used for all three arrays.
- Small asymmetries remain: measured channels sit $\sim$10–20 ns later after alignment than simulated ones (bounded at $\sim$0.5 dB of the CDL-vs-pool gap), and the correlation between the two antenna rows is lower in the measured data than in either model, for reasons not established.
- Most randomly placed users in the rooftop simulations received no paths and were dropped, so simulated users skew toward simpler links.
- The scattering result uses one scattering value and 7 cities, and was chosen after seeing the dry run; matching the users' distance undid most of it.

A code review during the measured-channel work found a cache keyed on object identity that could hand a scene another scene's ground mesh; it was fixed and everything rerun, and reported numbers moved by at most 0.02 dB.

7 Conclusion

In simulation, a covariance prior pooled over many ray-traced cities looked like a good starting point for an unseen site and a safer one than any single city. On measured channels from one site, it was worse than the 3GPP CDL models, because the simulated channels were too clean; with diffuse scattering and diffraction enabled it roughly matched CDL, but only with users placed farther away than the real ones. Learned models never beat the classical methods; at best they refined them. For a ray-traced prior to be useful at a real site, the simulator's settings matter at least as much as the number of cities.

Where next. Three directions follow from the results. (i) *Tune the simulator, not the covariance, from site measurements:* the gap is driven by global simulator settings and missing scene content, so a few site measurements could set the scattering level (or a clutter model) that is then used to simulate other cities, with no 3D model of the target site. Existing calibration work tunes a single, fully modelled site [12]; a coarse grid over the scattering coefficient, scored on 50 site measurements, is a cheap first test. (ii) *More measured sites*, since one street cannot settle a question about unseen sites; we found no public outdoor dataset with several sites and a planar array. (iii) *Prediction on measured channels*, where the classical method's advantage rests on a simulator that matches its own model; this needs millisecond-spaced measurements of moving users.

Reproducibility. Code, the lab notebook with every pre-registration and result in order, logs and result tables accompany the paper. Channel sets are regenerated by the scripts; the measured data is public under CC BY 4.0 [6].

References

[1] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," TR 38.901.

[2] J. Hoydis, S. Cammerer, F. Ait Aoudia, A. Vem, N. Binder, G. Marcus, A. Keller, "Sionna: An open-source library for next-generation physical layer research," arXiv:2203.11854, 2022.

[3] J. Hoydis, F. Aït Aoudia, S. Cammerer, M. Nimier-David, N. Binder, G. Marcus, A. Keller, "Sionna RT: Differentiable ray tracing for radio propagation modeling," arXiv:2303.11103, 2023.

[4] Functions Lab, "geo2sigmap," `https://github.com/functions-lab/geo2sigmap` (commit 08ca016).

[5] F. Euchner, M. Gauger, S. Dörner, S. ten Brink, "A distributed massive MIMO channel sounder for 'big CSI data'-driven machine learning," WSA 2021; arXiv:2206.15302.

[6] University of Stuttgart, "CSI dataset dichasus-dxxx-complete: Outdoor—three arrays distributed along the facade," DaRUS, doi:10.18419/DARUS-3153, CC BY 4.0.

[7] L. Del Moro *et al.*, "Site geometry and calibration uncertainties in digital twin-enabled channel estimation," arXiv:2607.09334, 2026.

[8] S. Alikhani *et al.*, "Digital twin aided channel estimation: Zone-specific subspace prediction and calibration," arXiv:2501.02758, 2025.

[9] M. S. Abouamer *et al.*, "Prediction of wireless channel statistics with ray tracing and uncalibrated digital twin," arXiv:2411.13360, 2024.

[10] C.-J. Zhao *et al.*, "SiFo: Wireless foundation model for low-overhead site-specific CSI feedback," arXiv:2605.16141, 2026.

[11] B. Liu *et al.*, "WiFo: Wireless foundation model for channel prediction," arXiv:2412.08908, 2024.

[12] J. Hoydis, F. Aït Aoudia, S. Cammerer, F. Euchner, M. Nimier-David, S. ten Brink, A. Keller, "Learning radio environments by differentiable ray tracing," arXiv:2311.18558, 2023.

[13] J. Morais, A. Malhotra, S. Hamidi-Rad, A. Alkhateeb, "Comparing stochastic and ray-tracing datasets in machine learning for wireless applications," arXiv:2512.12449, 2025.